



برنامج الدبلوم التخصصي  
Specialized Diploma Program  
Technical Tracks Details



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## Program Design

The specialized Diploma is a 36 Weeks long diploma, 30 Weeks of guided studies and 6 Weeks practical training. The load is 37 Hours/Week which is equivalent to 1330 Hours (1110 Guided Studies + 220 Practical). The diploma includes the following components:

### Technical Components

- 1- Academic (450 Hours)
- 2- Professional & Self-Learning (300 Hours)
- 3- Practical (220 Hours)

### Non-Technical Components

- 4- Soft Skills (120 Hours)
- 5- Leadership (120 Hours)
- 6- Languages (120 Hours)
- 7- Self-Improvement (30 Hours)

The Technical Components are designed to accommodate in the following tracks:

- Artificial Intelligence and Data Science
- Embedded Systems
- Cybersecurity
- AI-based software development
- Networks & IT Infrastructure
- Applied Artificial Intelligence and Data Analytics

The following sections describe the details of these technical Tracks.



# Artificial Intelligence and Data Science

## Target Students

- Graduates of the College of Engineering (Departments of Communications and Electronics, Department of Computer Engineering, Department of Biomedical Engineering, Department of Mechatronics)
- Graduates of the College of Computer Science and Information / College of Artificial Intelligence
- Graduates of the College of Science (Department of Computer Science)
- Graduates of the College of Commerce (Department of Business Information Systems)

## Program Objectives

- Provide Foundational and Advanced Knowledge in Artificial Intelligence and Data Science.
- Develop Practical Skills for Real-World AI Challenges.
- Foster Critical Thinking and Problem-Solving Skills.
- Prepare Students for AI Algorithms.
- Enable Students to Apply Knowledge through a Capstone Project.

## Academic Courses

- Course 1: Mathematical Foundations and Software Development for AI
- Course 2: Data Analysis & Engineering
- Course 3: Artificial Intelligence Concepts and Machine Learning Techniques
- Course 4: Deep Learning Techniques and Computer vision
- Course 5: Natural Language Processing
- Industry Project in Artificial Intelligence



# Embedded Systems

## Target Students

- Graduates of the College of Engineering (Departments of Communications and Electronics, Department of Computer Engineering, Department of Biomedical Engineering, Department of Mechatronics)

## Program Objectives

- Basic and Advanced knowledge in Embedded Systems.
- Practical and industrial skills for Embedded Systems.
- Problem-Solving Skills.
- Embedded Systems Engineer job profile building.
- Capstone project for real project implementation.

## Academic Courses

- Course 1: Embedded-C Programming.
- Course 2: Microcontroller Interfacing Level 1.
- Course 3: Microcontroller Interfacing Level 2.
- Course 4: Real Time Operating Systems (RTOS).
- Course 5: Smart Embedded Systems and Applications
- Course 6: Industrial Project in Embedded Systems.



# Cybersecurity

## Target Students

- Graduates of the College of Engineering (Departments of Communications and Electronics, Department of Computer Engineering, Department of Biomedical Engineering, Department of Mechatronics)
- Graduates of the College of Computer Science and Information / College of Artificial Intelligence
- Graduates of the College of Science (Department of Computer Science)

## Program Objectives

- Provide Foundational and Advanced Knowledge in Cybersecurity.
- Develop Practical Skills for Real-World Cybersecurity Challenges.
- Foster Critical Thinking and Problem-Solving Skills.
- Prepare Students for Cybersecurity Roles in Modern Organizations.
- Enable Students to Apply Knowledge Through a Capstone Project.

## Academic Courses

- Course 1: Foundations of Cybersecurity
- Course 2: Network Security and Cyber Defense
- Course 3: Ethical Hacking and Penetration Testing
- Course 4: Incident Response and Digital Forensics
- Course 5: Cybersecurity Risk Management and Compliance
- Cybersecurity Project



# AI-based software development

## Target Students

- Graduates of the College of Engineering (Departments of Communications and Electronics, Department of Computer Engineering, Department of Biomedical Engineering, Department of Mechatronics)
- Graduates of the College of Computer Science Information and Artificial Intelligence Colleges
- Graduates of the College of Science (Computer Science Department)
- Graduates of any university (Subject to passing the entry exam)

## Program Objectives

- Develop Full-Stack Software Development Skills
- Master Software Engineering Best Practices
- Enhance Development Efficiency with AI-Powered Tools
- Gain Expertise in Cloud Computing & DevOps
- Implement Secure Coding and Cybersecurity Practices
- Optimize Software Performance & Scalability
- Apply Knowledge in a Real-World Capstone Project

## Academic Courses:

- Course 1: Foundations of Software Development
- Course 2: Frontend Development & User Experience
- Course 3: Backend Development & Databases
- Course 4: DevOps, Cloud Computing & AI-Driven Automation
- Course 5: Advanced Software Development & Security
- Final Project: Full-Stack Software Development Capstone



# Networks & IT Infrastructure

## Target Students

- Graduates of the College of Engineering (Departments of Communications and Electronics, Department of Computer Engineering, Department of Biomedical Engineering, Department of Mechatronics)
- Graduates of the College of Computer Science Information and Artificial Intelligence Colleges

## Program Objectives

- Provide Foundational and Advanced Knowledge in Networks & IT Infrastructure.
- Develop Practical Skills for Real-World Networks & IT Infrastructure Challenges.
- Foster Critical Thinking and Problem-Solving Skills in Networks multi-infrastructure domain.
- Prepare Students for modern Networks Design, Architectures, Services, and Operations.
- Enable Students to Apply Knowledge Through a Capstone Project.

## Academic Courses

- Course 1: Networking Fundamentals.
- Course 2: Enterprise Networking and Automation.
- Course 3: Server Fundamentals: Linux and Windows.
- Course 4: Wireless Networking Fundamentals.
- Course 5: Network security
- Course 6: Industrial Project in Networks & IT Infrastructure.



# Applied AI and Data Analytics

## Target Students

- Graduates' students from any specialization

## Program Objectives

- Apply mathematical, statistical, analytical, and machine learning/deep learning techniques to gain insights from large and complex datasets.
- Tackle real-world challenges in data science, from optimizing databases to analyzing social trends and transforming raw data into strategic solutions.
- Design predictive models.
- Foster Critical Thinking and Problem-Solving Skills.
- Apply knowledge through a Capstone Project.

## Academic Courses

- Course 1: Statistics and Probabilities for Data Analytics.
- Course 2: Databases.
- Course 3: Data Mining and Visualization.
- Course 4: Practical Machine and Deep learning.
- Course 5: Web and Social Networks Analytics
- Course 6: Capstone project for real problem.